STEP 7 OF 8

Structural Design and Thermal Analysis

Chassis Design · Mass Budget · Vibration · Thermal Control · Materials

Al 6061-T6 Chassis | 2,445 g Total Mass | 215 g Margin (8.1%) | Passive Thermal | CDS Rev 14 Compliant | No Moving Parts

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OFFICIAL MISSION PARTNER CO-DESIGN DOCUMENT

MTQ-Only Revision | Reaction Wheel Removed | 3-Axis Magnetorquer System

1 Structural and Thermal Overview

The SLARS-BCG-1 structure performs four simultaneous functions: providing the rigid mechanical frame that survives launch loads; housing and aligning every subsystem; conducting waste heat to the external panels for radiation to space; and shielding sensitive components from radiation. The key change from removing the reaction wheel is an 80 g mass saving, freed volume in the bottom bus section, and elimination of RW-induced micro-vibration — a direct image quality benefit.

Al 6061-T6	~2,445 g	~215 g	8.1%	Passive
Chassis material	Total mass	Mass margin	Margin percentage	Thermal control

MTQ-ONLY STRUCTURAL CHANGES

ADCS mass: 195 g -> 115 g (80 g reduction — reaction wheel removed)
Volume freed: 40 x 40 x 20 mm in bottom bus section (RW mounting footprint)
Reaction wheel vibration isolation mount (neoprene grommets): REMOVED
Reaction wheel thermal heater (0.05 W): REMOVED — thermal budget improved
Total spacecraft mass: 2,525 g -> 2,445 g | Margin: 135 g -> 215 g (5.1% -> 8.1%)
Micro-vibration: ELIMINATED — no spinning components — direct image quality benefit

2 Inputs from Steps 1–6 — Structural Requirements

Source	Structural / Thermal Requirement	Impact
Step 1	500 km LEO: AO exposure, radiation, thermal cycling	Anodised Al panels; component shielding
Step 2	Battery must stay above -10 deg C	Battery heater 0.1 W; position near centre of mass
Step 2	Solar cells on all 4 long faces	GaAs cell adhesive bonding to Al panels
Step 3	Imager nadir-facing at top of stack	Top face: 25 mm clear aperture in panel
Step 3	SDR + AIS board: 22x90x90 mm	PCB slot allocated in mid-stack position
Step 4	UHF whip deploys from bottom	Spring-loaded monopole retention mechanism
Step 4	S-band patch on top face: 62x62 mm	Top panel flush-mounted patch antenna
Step 5 (MTQ)	MTQ rods: 3x 90 mm along body edges	Rods bonded in chassis corner grooves
Step 5 (MTQ)	NO reaction wheel — mount removed	Bottom bus zone: 40x40x20 mm volume freed
Step 6	PC/104 stack: 15 mm pitch between boards	Chassis PC/104 standoff spacing: 15 mm +/-0.1 mm
Launch std	CDS Rev 14: 100x100x227 mm envelope	Rails 8.5 mm min; no protrusion beyond envelope
Launch std	Random vibration: 14.1 Grms (GEVS)	Structure must survive GEVS-7000 levels

3 Complete Mass Budget — All Subsystems (MTQ-Only Updated)

Subsystem	Components	Mass (g)	% Total	Change vs Previous
Structure and chassis	Rails, panels, caps, standoffs, fasteners	390	15.9%	Unchanged
EPS and power	GomSpace P31u EPS board + 2S2P Li-ion battery	345	14.1%	Unchanged
Payload — Imager	Multispectral camera + optics + filter	205	8.4%	Unchanged
COMMS	UHF transceiver + S-band TX + antennas + diplexer	225	9.2%	Unchanged
ADCS — MTQ only	3x MTQ rods + sensors + ADCS PCB (no RW	)115	4.7%	**-80 g (RW removed)**
OBC	ARM Cortex-M4 flight computer (PC/104)	80	3.3%	Unchanged
Payload — SDR + AIS	Combined RF payload board + RF components	85	3.5%	Unchanged
Solar cells	GaAs cells on 4 faces + adhesive + wiring	40	1.6%	Unchanged
Thermal control	Battery heater PCB + MLI (RW heater removed)	35	1.4%	-5 g (RW heater out)
Harness	Internal wiring, connectors, PC/104 cables	80	3.3%	Unchanged
Mass margin (8.1%)	Design reserve — unallocated	215	8.8%	+80 g improvement
TOTAL	**All subsystems + margin**	**2,445**	**100%**	**2,525 -> 2,445 g**

MASS BUDGET STATUS: 2,445 g of 2,660 g maximum. Margin: 215 g (8.1%).
This exceeds the standard 5% preliminary design target — comfortable margin for a first satellite.
Removing the reaction wheel improved margin from 135 g (5.1%) to 215 g (8.1%).

4 Stack Layout — Physical Configuration (MTQ-Only)

Position (Top to Bottom)	Component	Height	Notes
Top cap	Antenna deployment + kill switch	14 mm	Release pin; UHF+VHF whips stowed
Payload zone 1	Multispectral imager module	45 mm	Nadir-facing lens at top face
Payload zone 2	Combined SDR + AIS RF board	22 mm	Shared antenna feed via coax
Payload zone 3	Volume margin / future payload	47 mm	Reserved — structural filler if unused
Bus zone 1	EPS board + battery pack	55 mm	2S2P Li-ion + MPPT PCB
Bus zone 2	OBC + COMMS radio stack	30 mm	PC/104 stack; ARM flight computer
Bottom cap	ADCS sensors + MTQ rods ONLY	14 mm	IMU, magnetometer, 3-axis coil rods — no RW
REMOVED	Reaction wheel assembly	(was 20 mm)	Not present — volume freed for harness routing

5 Chassis Design — Al 6061-T6 CDS Rev 14

Load Case	Level	Axis	Design Factor
Quasi-static acceleration (axial)	25.0 g	Z (along rail)	1.25x factor of safety
Quasi-static acceleration (lateral)	15.0 g	X and Y	1.25x factor of safety
Random vibration (all axes)	14.1 Grms overall	X, Y, Z — 120 s each	Fatigue life analysis
Acoustic pressure	140 dB SPL overall	All surfaces — 120 s	Panel skin analysis
Shock (separation)	300 g SRS at 10 kHz	All axes	

Component	Dimensions	Material	Mass	Function
Main rails (x2)	8.5x8.5x227 mm L-section	Al 6061-T6	~45 g each	CDS slide rails; primary load path
Side panels (x4)	100x227x1.5 mm	Al 6061-T6 anodised	~40 g each	Solar substrate; radiation shield; radiator
Top cap	100x100x6 mm	Al 6061-T6	~25 g	S-band patch; imager port; kill switch
Bottom cap	100x100x6 mm	Al 6061-T6	~25 g	UHF antenna; burn resistor; separation switch
MTQ grooves	3x corner grooves, 8 mm wide	Machined into rails	—	Houses MTQ rod along each axis
REMOVED: RW mount	40x40x3 mm plate	N/A	Saves ~8 g	Reaction wheel isolation mount — removed
Total chassis	Assembled 2U frame	All Al 6061-T6	~390 g	15.9% of 2,660 g total budget

6 Launch Loads and Vibration

Single event — fragile components assessed

Analysis Result	Value	Assessment	Mitigation
Rail axial stress at 25 g	~12 MPa	Al 6061-T6 yield 276 MPa — FoS = 23	No action needed
Panel bending stress	~8 MPa	FoS = 35 — panels over-designed	May thin to 1.0 mm if mass critical
PC/104 standoff shear at 25 g	~3.2 MPa	Al shear yield ~150 MPa — FoS = 47	M2 fasteners + Loctite 243
RW vibration isolation	REMOVED — not applicable	**No spinning component**	No neoprene grommets needed
Micro-vibration from ADCS	NONE — MTQ-only	**Image quality benefit vs RW design**	Inherent MTQ advantage
First structural mode (est)	~180 Hz	Above 100 Hz GEVS target — compliant	Confirm with FEA before PDR

7 Thermal Design — Passive Architecture

Total TCS	All passive elements combined	No pumps, fans, or heat pipes	~35 g

Thermal control is entirely passive: surface coatings + chassis conduction + two resistive heaters (battery + RW heater removed). The RW motor heat source (0.15 W avg) is eliminated. Average total dissipation is now ~1.35 W — well within the

TCS Element	Implementation	Thermal Function	Mass
Long face panels	Hard anodised Al (black, emissivity 0.88)	Radiates IR to space	< 1 g (surface
Top panel (nadir)	White epoxy paint (Z-306, emissivity 0.92)	Reflects solar; cools imager + S-band	< 1 g
Bottom panel	Bare Al mill finish (emissivity 0.05)	Reduces radiation loss in eclipse	< 1 g
Chassis rails	Bare Al 6061 machined	Primary conduction path: PCB to panel	Structure
PCB thermal interface	Copper pour contacting chassis rails	Transfers 0.5 W OBC heat per board	< 2 g
Battery heater	10 ohm resistor PCB, 0.1 W, -5 deg C thermostat	Prevents Li-ion below -10 deg C	~3 g
Battery MLI blanket	2-layer double-aluminised mylar	Reduces thermal cycling amplitude	~5 g
REMOVED: RW heater	5 ohm resistor at bearing, 0.05 W	Not applicable — no reaction wheel	Saves 2 g

~19.5 W radiative capacity.

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8 Component Temperature Limits and Predicted Ranges

Component	Survival Range	Operating Range	Predicted Hot	Predicted Cold	Status
Li-ion battery	-40 to +60 C	-10 to +45 C	+32 C	-7 C (heater on)	OK with heater
OBC ARM M4	-40 to +85 C	-20 to +70 C	+55 C	+5 C	OK — margin
GaAs solar cells	-100 to +100 C	-40 to +80 C	+72 C	-85 C (eclipse)	OK — within spec
Multispectral imager	-30 to +85 C	0 to +60 C	+42 C	+8 C	OK — monitor hot
UHF radio	-40 to +85 C	-20 to +70 C	+48 C	+2 C	OK — margin
S-band TX	-30 to +80 C	-10 to +65 C	+55 C	+4 C	Monitor hot case
ADCS MTQ rods	-40 to +85 C	-40 to +85 C	+50 C	-25 C	OK — wide range
REMOVED: RW bearing	N/A	N/A	N/A	N/A	Not present
EPS board	-40 to +85 C	-20 to +70 C	+45 C	+2 C	OK — margin
eMMC flash	-40 to +85 C	-25 to +85 C	+50 C	-5 C	OK — wide range

9 CDS Rev 14 Compliance Checklist

CDS Rev 14 Requirement	SLARS-BCG-1 Compliance	Status
External dimensions 100x100x226.3 mm	Chassis: 100x100x227 mm machined	PASS
Rail width 8.5x8.5 mm min	Rails machined 8.5x8.5 mm — confirmed	PASS
No protrusions beyond envelope	Antennas stowed; patch flush to panel	**PASS**
Antenna deployment mechanism	Dual nylon burn wire for UHF+VHF monopoles	**PASS**
Kill switches (x2)	2x mechanical switches on rail tabs	**PASS**
Material outgassing TML < 1%	All adhesives + coatings screened — bake-out required	
Magnetic cleanliness < 0.1 A.m2	MTQs de-energised at deploy; PCB loops cancelled — est 0.01A.m	**PASS**2
Centre of mass < 2 mm offset	CoM ~0.5 mm offset (improved vs RW version)	**PASS**
Vibration survival GEVS-7000	Analytically compliant — physical test required	**PENDING TEST**
RF transmit inhibit	Kill switches prevent all TX — confirmed in OBSW	PASS
		PASS after bake-out

Locked Parameter	Value (MTQ-Only)	Notes
Chassis material	Aluminium 6061-T6, hard anodised	Heritage on >1,500 CubeSat mission
Total spacecraft mass	~2,445 g (of 2,660 g max)	Launch vehicle mass requirement confirmed

10 Step 7 Locked Output Parameters — Feeds Step 8

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Mass margin	~215 g (8.1%)	Improved from 135 g (5.1%) by removing RW
External dimensions	100x100x227 mm	P-POD deployer confirmed — standard 2U
CDS compliance	PASS on all static requirements	Vibration test is remaining launch gate
First structural mode	~180 Hz estimated	Above LV first mode — confirm with FEA
Surface finish — long faces	Black anodise + GaAs; alpha 0.92, eps 0.88	Thermal model confirmed
Surface finish — top (nadir)	White paint Z-306; alpha 0.15, eps 0.92	Keeps imager + S-band cool
Thermal control type	Passive — conduction + radiation + 1 heater	No active TCS — no power allocation
Battery heater	0.1 W at -5 deg C thermostat setpoint	Within EPS heater allocation
RW heater	REMOVED — no reaction wheel	Thermal budget improved by 0.05 W
Micro-vibration	NONE — MTQ-only has no moving parts	Direct image quality benefit

Battery temperature	Predicted -7 to +32 C (with heater)	Within -10 to +45 C operating range
OBC temperature	Predicted +5 to +55 C	Within -20 to +70 C operating range
Vibration test required	GEVS-7000 before launch — mandatory	Contact NASTEC or ISRO SAC facility
Kill switches	2x mechanical on rail tabs	Step 8: tested on pad before encapsulation
AIT duration	~24 weeks (1 week shorter — no RW integration)	Step 8: major schedule driver

STEP 7 LOCKED — Reaction wheel removed. ADCS mass 195 g -> 115 g. Total mass 2,525 g -> 2,445 g. Margin 215 g